



Service Tool Instruction

Service Tool Instruction
Number: 3377926

Revision Level:

Released Date: 11-dec-
2002

Description

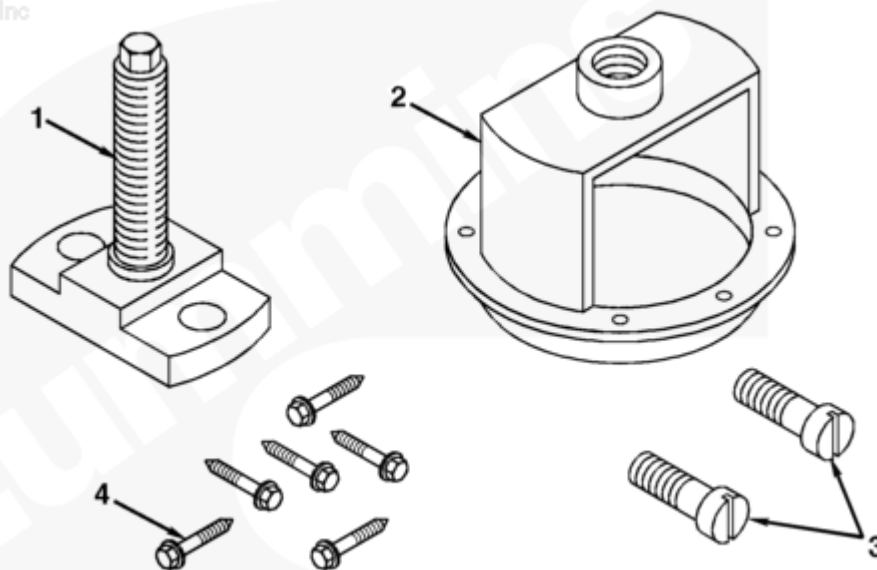
Front Crankshaft Seal Replacer Kit

Purpose

This service tool instruction explains the use of the front crankshaft seal replacer kit, Part Number 3164926. The front crankshaft seal replacer kit is used to remove and install the QSK23 engine front crankshaft seal and the dust seal (used on construction applications).

Note : A new dust seal is not included with a new front crankshaft seal and must be ordered separately.

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Table 1, Front Crankshaft Seal Replacer Kit, Part Number 3164926

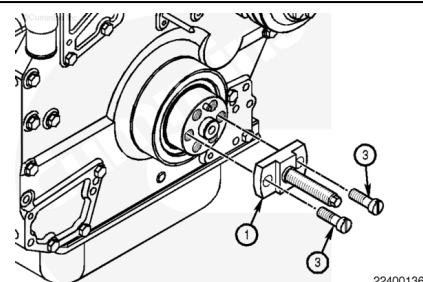
Item Number	Part Number	Description	Quantity
1	3164972	Replacer screw assembly	1
2	3164968	Crankshaft seal replacer	1
3	3163489	Round head cap screw, M22 x 1.5 x 40-mm	2

Table 1, Front Crankshaft Seal Replacer Kit, Part Number 3164926

Item Number	Part Number	Description	Quantity
4	3163828	Sheet metal screw, number 10 x 50.8-mm [2-in] (6 shown)	36
Not shown	3164218	Drill, [9/64-in]	1

Mount the replacer screw assembly (1) onto the crankshaft nose.

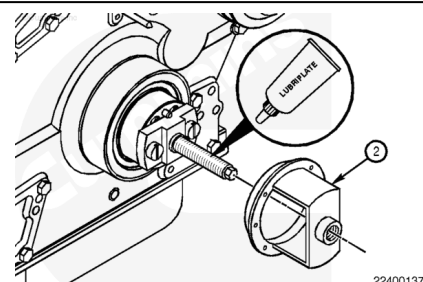
Install the M22 x 1.5 x 60-mm round head capscrews (3).



Lubricate the replacer screw assembly with Lubriplate™, Part Number 3163086, Part Number 3163087 or a suitable grease.

Hold the replacer screw and install the crankshaft seal replacer (2) onto the replacer screw assembly.

Advance the crankshaft seal replacer toward the front seal or dust seal by rotating it **clockwise** until it is positioned against the front seal or dust seal.



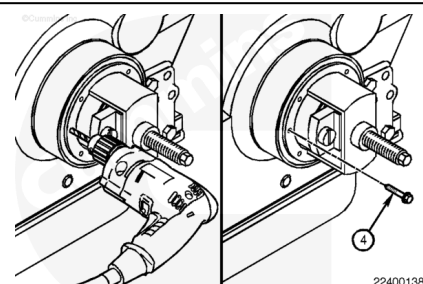
⚠ CAUTION ⚠

Drill holes carefully and straight to reduce the possibility of damage to the front cover or the crankshaft.

For in chassis use, if space is restricted, use a compact right angle drill or shorten the drill bit as necessary.

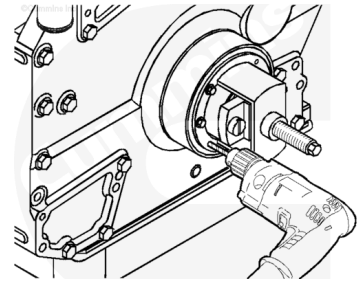
Apply grease to the drill to catch the chips. Stop frequently to remove the chips. With the crankshaft seal replacer against the front seal or (dust seal on construction applications), drill one hole 10-mm [0.39 in] deep through the oil seal casing.

Install one sheet metal screw into the front seal to hold the crankshaft seal replacer in position.



Drill five additional holes 10-mm [0.39] deep and install the sheet metal screws.

All six sheet metal screws **must** be threaded through both the dust seal (if equipped) and front seal. The sheet metal screws need to be uniformly tightened in order to pull out both seals together.

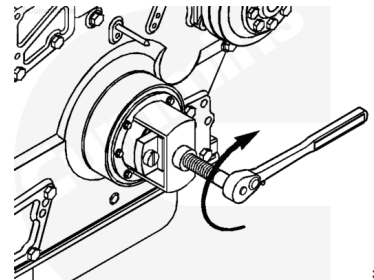


⚠ CAUTION ⚠

Do not use an impact wrench or air tools. Doing so can damage the tool.

Slowly rotate the replacer screw **clockwise** until the seals are removed. Use a ratcheting box wrench such as Snap-On, Part Number RBLM1315, or equivalent to facilitate the removal and installation of the seal in a restricted space.

If the sheet metal screws pull out of the seals, stop the removal operation. Remove the sheet metal screws, reorient the crankshaft seal replacer, drill new holes, and reinstall the sheet metal screws. Complete the removal procedure, remove the replacer, and discard the oil seal.



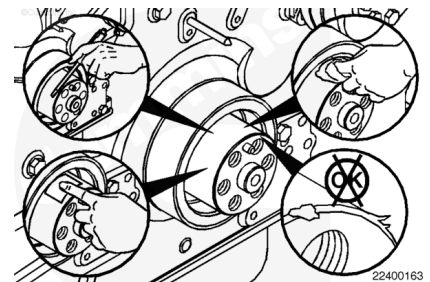
Before installing both front seals, make sure the metal chips are completely removed.

Use a magnet to remove any metal chips from crankshaft.

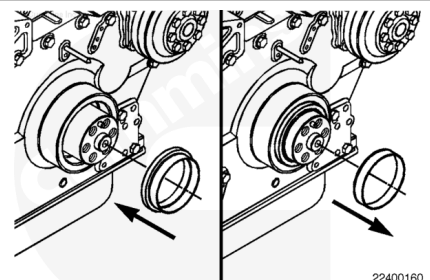
Use a crocus cloth to polish and remove any deposits from the crankshaft. Visible scratches on the sealing surface can result in seal leakage.

Use a clean, lint-free cloth to clean the crankshaft flange and remove all traces of oil film.

Check crankshaft end for nicks, burrs, and grooves.

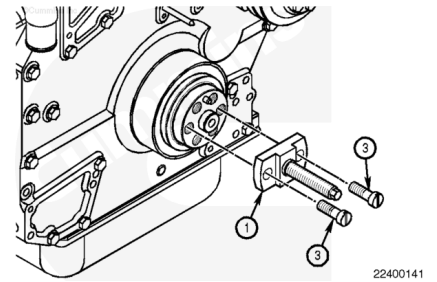


Using the plastic installation sleeve, place the new oil seal over the crankshaft nose and slide the seal toward the front cover. Make sure the oil seal is positioned squarely on the crankshaft. Avoid cocking the seal. Remove the plastic installation sleeve supplied with the oil seal.



Mount the replacer screw assembly (1) onto the crankshaft nose.

Install the M22 x 1.5 x 60-mm round head capscrews (3).

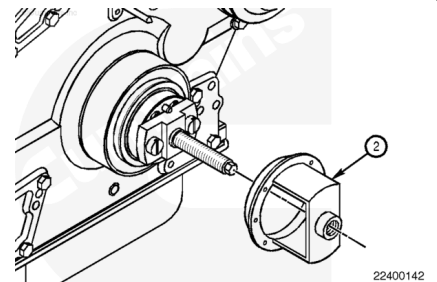


⚠ CAUTION ⚠

Do not use an impact wrench or air tools. Doing so can damage the tool.

Hold the replacer screw and install the crankshaft seal replacer (2) onto the replacer screw assembly.

Advance the crankshaft seal replacer toward the front seal by rotating it **clockwise** until it is positioned against the front seal.

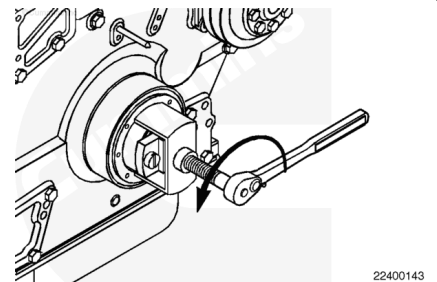


⚠ CAUTION ⚠

Do not overtighten the replacer screw assembly after the crankshaft seal replacer contacts the front cover. Doing so can damage the tool.

Holding the crankshaft seal replacer, rotate the replacer screw **counterclockwise** until the crankshaft seal replacer is positioned against the front cover.

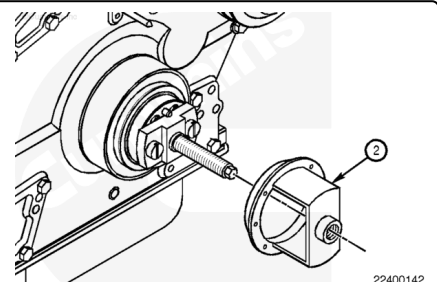
Remove the crankshaft seal replacer from the replacer screw assembly.



Note : The dust seal must be ordered separate from the front seal.

Place the dust seal (used on construction applications) over the crankshaft nose and slide toward the front cover. Make sure the dust seal is positioned squarely on the crankshaft. Avoid cocking the dust seal.

Hold the replacer screw and install the crankshaft seal replacer (2) onto the replacer screw assembly. Advance the crankshaft seal replacer toward the dust seal by rotating it **clockwise** until it is positioned against the dust seal.



Holding the crankshaft seal replacer, rotate the replacer screw **counterclockwise** until the dust seal is positioned to the required depth.

An under cut (1) on the crankshaft seal replacer will determine the correct depth (4-mm) of the dust seal. Do **not** contact the crankshaft seal replacer against the front cover.

Remove the front crankshaft seal replacer and the replacer screw assembly.

